

Rishabh Tennankore, Ph.D.

[Portfolio](#) | [Google Scholar](#) | [Email](#)

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PROFESSIONAL PROFILE

Materials Science Ph.D. with hands-on experience in cleanroom fabrication, vacuum-based thin-film processing, and process troubleshooting for nanoscale systems. Experienced developing and optimizing multi-step fabrication workflows involving PVD, lithography, dry etch, surface treatment, and process metrology to improve robustness, pattern fidelity, and manufacturing yield.

SKILLS

Cleanroom Experience

- *Fabrication & Processing*
 - Lithography (Photo, Nano-Imprint), Dry Etch (Plasma, Ion Beam), PVD (e-beam evaporation, sputtering), CVD (vapor-deposition of silanes; mTorr regime), Wet processing, Substrate handling
- *Metrology & Characterization*
 - SEM (Top down, X-section), FIB-SEM, EDS, Optical Microscopy, Laser Confocal Microscopy, Ellipsometry, Profilometry, Contact Angle, UV/Vis Spectroscopy

Process Reliability

- Statistical Process Control (SPC), Root-Cause Analysis (RCA), Design of Experiment (DOE), Recipe development, Failure Analysis, Process Documentation

Modeling & Data Analysis

- Python, MATLAB, JMP, LabVIEW, MS Excel, AutoCAD, ImageJ

WORK EXPERIENCE

Process Engineer

May 2026

Meta Reality Labs (Contract through HCLTech) – Redmond, WA

Supporting nanofabrication operations for Meta Reality Labs' augmented reality platforms within an R&D cleanroom environment, owning lithography, dry etch and destructive metrology process modules.

- Supporting advanced manufacturing of key enablers of next-gen AR displays

Membrane Development Scientist

Oct 2025 – Dec 2025

OsmoPure Technologies Inc. – Boulder, CO

Owned end-to-end thin-film process module including coating, curing, substrate integration, metrology, and defect troubleshooting in a pilot manufacturing environment.

- Executed **100+ process iterations** to establish stable operating windows, troubleshoot variability, and improve reproducibility across multiple material platforms
- Performed **SEM-driven root cause analysis** on coating failures and implemented corrective actions that **reduced defect rates by >90%**
- **Reduced thickness variation from ~20% to <3%** through standardized metrology workflows, process controls, and operator repeatability improvements

Graduate Research Assistant

Aug 2020 – Aug 2025

University of Colorado, Materials Science and Engineering – Boulder, CO

Operated multi-step cleanroom fabrication workflows involving photolithography, metal deposition (e-beam evaporation/sputtering), surface modification, and device integration.

- Developed and executed **multi-step fabrication workflows** integrating **photolithography, PVD** (SU-8 patterning + 10 nm Au) and surface functionalization via **CVD** focusing on process repeatability, defect mitigation, and interface control
- **Optimized e-beam and sputtering deposition parameters** to achieve conformal coatings on high-aspect-ratio structures; reduced pattern defects and improved uniformity
- **Deposited and characterized ~100 nm functional thin-films** on porous substrates, achieving greater than 100× transport property improvement through thickness and composition control

R&D Engineer

Jan 2018 – Dec 2018

BASF – Wyandotte, MI

- **Led cross-functional effort to design and validate methods** to test shear-driven performance across 4 surfactant formulations, established structure–property–performance relationships – informing downstream product development decisions
- Applied CFD modeling, statistical analysis and machine learning methods to **develop a regression model** validating parameter–response correlations and **achieved adjusted R² up to 0.97**

EDUCATION

Doctor of Philosophy (Ph.D.), Materials Science and Engineering

Aug 2020 – Aug 2025

- University of Colorado Boulder; GPA: 3.96 / 4.00

Master of Science (M.S.), Macromolecular Science and Engineering

Sep 2017 – Dec 2018

- University of Michigan, Ann Arbor; GPA: 3.96 / 4.00

PUBLICATIONS

5 peer-reviewed journal articles + 1 article submitted